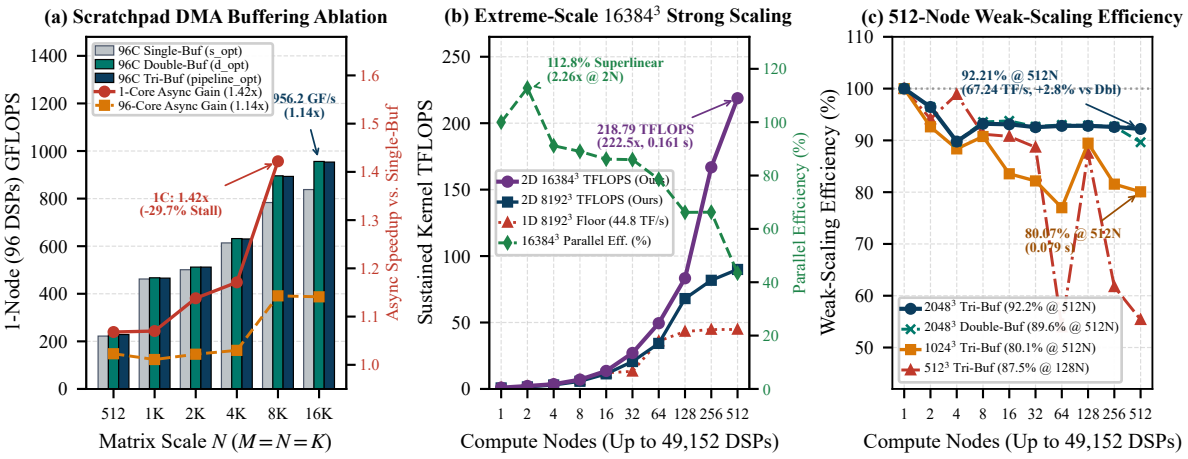

(a) Scratchpad DMA Buffering Ablation
96C Single-Buf (s_opt)
96C Double-Buf (d_opt)
96C Tri-Buf (pipeline_opt)
1-Core Async Gain (1.42x)
96-Core Async Gain (1.14x)
956.2 GF/s (1.14x)
1C: 1.42x (-29.7% Stall)
1-Node (96 DSPs) GFLOPS
Async Speedup vs. Single-Buf
Matrix Scale N (M = N = K)
(b) Extreme-Scale 16384³ Strong Scaling
112.8% Superlinear (2.26x @ 2N)
218.79 TFLOPS (222.5x, 0.161 s)
2D 16384³ TFLOPS (Ours)
2D 8192³ TFLOPS (Ours)
1D 8192³ Floor (44.8 TF/s)
16384³ Parallel Eff. (%)
Sustained Kernel TFLOPS
Parallel Efficiency (%)
Compute Nodes (Up to 49,152 DSPs)
(c) 512-Node Weak-Scaling Efficiency
92.21% @ 512N (67.24 TF/s, +2.8% vs Dbl)
80.07% @ 512N (0.079 s)
2048³ Tri-Buf (92.2% @ 512N)
2048³ Double-Buf (89.6% @ 512N)
1024³ Tri-Buf (80.1% @ 512N)
512³ Tri-Buf (87.5% @ 128N)
Weak-Scaling Efficiency (%)
Compute Nodes (Up to 49,152 DSPs)